Interpretable Cross-Sensor Hyperspectral Image Super-Resolution

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***Abstract*—Hyperspectral image super-resolution (HSI SR) has attracted increasing research attention. Despite recent advances, existing HSI SR methods face two major limitations: 1) differences across sensors hinder the generalization of models trained on source-sensor data to unseen sensor data and 2) data-driven methods generally lack interpretability and do not sufficiently exploit the intrinsic spectral properties of HSIs. To address these limitations, we propose an interpretable cross-sensor HSI SR framework that integrates explicit spectral priors into an optimization-driven unfolding network. This method enables the trained model to be applied to unseen target-sensor data without additional retraining. Specifically, an endmember-aware dynamic modulation network (EDMN) is designed to encode compact transferable material-related priors from endmembers into adaptive bandwise weights to modulate iterative HSI SR reconstruction, improving cross-sensor generalization. Furthermore, a local spectral smoothness and spatial-gradient constraint (LSSSC) is incorporated into the unfolding network. By combining LSSSC with adaptive attention weights, the proposed network promotes spectral smoothness in homogeneous regions while mitigating excessive smoothing near boundaries, thereby improving spectral preservation. Cross-sensor experiments across three datasets show that the proposed method achieves competitive overall performance. In addition, downstream HSI classification is performed to verify model performance.**

***Index Terms*—Cross-sensor, hyperspectral image (HSI), interpretable network, spectral prior, super-resolution (SR).**

I. Introduction

HYPERSPECTRAL remote sensing captures reflectance data from terrestrial surfaces across hundreds of contiguous, narrow spectral bands, producing 3-D hyperspectral images (HSIs) [1], [2]. Each 3-D HSI comprises a stack of 2-D grayscale images containing the spatial–spectral features of the observed scenes, while the spectral signature of each pixel facilitates the precise identification of materials [3]. Due to these unique properties, HSIs are widely used for semantic segmentation [4], land cover classification [5], change detection [6], and water quality monitoring [7].

However, the tradeoff between spatial resolution and spectral fidelity restricts the spatial resolution of HSIs [8]. HSI super-resolution (SR) has emerged as a promising method for generating high-resolution HSIs from low-resolution HSIs. Current mainstream HSI SR methods can be classified as fusion-based HSI SR [9] and single HSI SR [10]. Fusion-based methods require a low-resolution HSI and a coregistered high-resolution auxiliary image (e.g., multispectral or panchromatic images) [11]. Nevertheless, misalignment due to discrepancies among sensors or scene dynamics limits the practical application of fusion-based methods. Therefore, single HSI SR has gained significant attention due to its flexibility in not requiring additional auxiliary images.

Despite remarkable progress in single HSI SR methods [10], [12], [13], [14], [15], [16], [17], the differences among HSIs from diverse sensors hinder model generalization. Specifically, as illustrated in Fig. 1(a), conventional methods are commonly trained and tested on data acquired by the same sensor, requiring separate training for each sensor. This paradigm limits their application to HSIs from unseen sensors. Recently, unified HSI processing has attracted increasing attention. These methods aim to build a single model for different scenes, degradations, or restoration tasks. For example, Wu et al. [18] proposed a multiprompt framework to achieve unified HSI restoration across diverse degradation types. In addition, a general HSI SR framework via meta-transfer learning has been proposed, which achieves better performance and generalizability [19]. Fig. 1(b) illustrates multisource training methods, which use multiple HSI datasets to train a single model. However, these methods rely on training with multiple datasets to improve generalization, limiting their practical applicability.

Another major limitation of current single HSI SR methods is their limited physical interpretability [20]. Deep learning-based HSI SR methods employ data-driven paradigms to achieve high-resolution reconstruction [13], [21]. However, these black-box models struggle to provide a clear physical interpretation of HSI SR [22]. To improve interpretability,

Received 27 July 2026; revised 20 August 2026; accepted 21 August 2026. Date of publication 26 August 2026; date of current version 14 September 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFD1700802; in part by the Key Research and Development Program of Hunan Province, China, under Grant 2023SK2041; in part by the Natural Science Foundation of Hunan Province, China, under Grant 2023JJ20018; and in part by Changsha Science and Technology Program under Grant kq2208036. *(Corresponding author: Bin Yang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3727427

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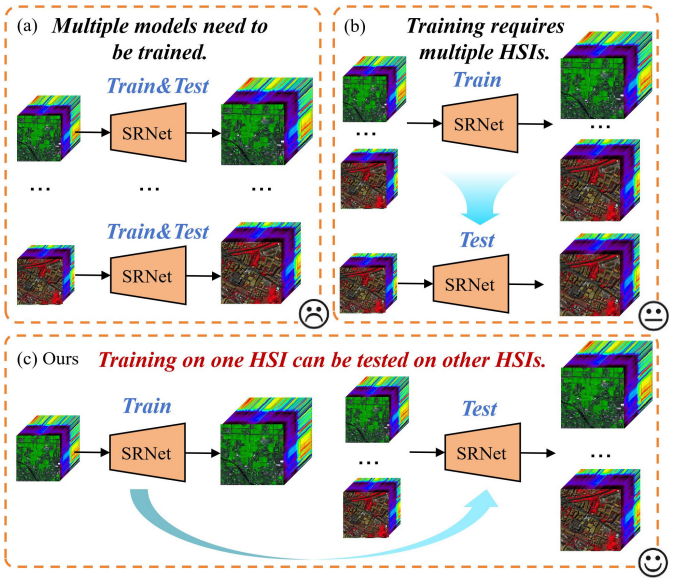


Fig. 1. Different types of HSI SR methods. (a) Simple HSI SR methods: each model is trained and tested on data from the same sensor, requiring separate models for different sensors. (b) Multisource training methods: a single model is trained on multiple HSI datasets, but its generalization to unseen sensors depends on the diversity of datasets. (c) Proposed HSI SR method: a model trained on source-sensor HSI can be applied to target-sensor HSIs without additional retraining.

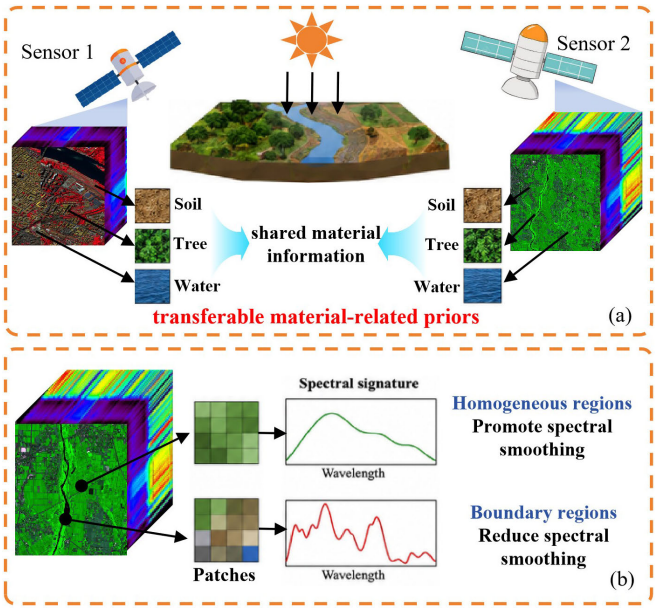


Fig. 2. (a) Illustration of transferable material-related priors across HSIs acquired by different sensors. (b) Different spectral smoothing demands in homogeneous and boundary regions.

optimization-driven methods, which combine model-driven formulations with data-driven learning, have been developed [12], [22], [23]. Nevertheless, existing optimization-driven methods mainly rely on neural networks to implicitly learn HSI priors and do not sufficiently incorporate explicit spectral properties. In particular, spectral signatures generally exhibit different smoothness characteristics in homogeneous and boundary regions, which should be considered during HSI SR. In summary, there are two major challenges.

1) *The Differences Among HSIs Acquired by Different Sensors Limit Model Generalization:* HSIs acquired by different sensors usually have different characteristics. However, most existing HSI SR methods are trained for a specific sensor and cannot be directly generalized to HSIs from unseen sensors, resulting in degraded cross-sensor HSI SR performance.
2) *Limited Model Interpretability and Insufficient Exploitation of Intrinsic Spectral Priors:* Although optimization-driven HSI SR methods improve interpretability through iterative unfolding, these methods still mainly rely on neural networks to implicitly learn HSI priors and insufficiently exploit intrinsic spectral properties, which may limit spectral preservation.

In this article, we propose an interpretable cross-sensor HSI SR framework with effective generalization performance, as illustrated in Fig. 1(c). Specifically, endmembers describe material-related spectral signatures that can provide relatively transferable information across different sensors, as shown in Fig. 2(a). This material-related information can serve as a transferable prior for improving cross-sensor generalization. Inspired by this, an endmember-aware dynamic modulation network (EDMN) is designed to encode compact transferable material-related priors into adaptive bandwise weights for iterative HSI SR reconstruction. In addition, spectral smoothness should not be applied equally to all regions because boundary or mixed pixels may have large spectral variations, as shown in Fig. 2(b). Therefore, an optimization-driven network with a local spectral smoothness and spatial-gradient constraint (LSSSC) is proposed to integrate explicit spectral–spatial gradient priors and adaptive attention weights, promoting spectral smoothness in homogeneous regions while alleviating smoothing near boundaries. The main contributions are summarized as follows.

1) An interpretable cross-sensor HSI SR framework is proposed to improve the generalization of a single model across different sensors. The trained model can be applied to unseen target-sensor HSIs without additional retraining.
2) An EDMN is designed to encode transferable material-related priors from endmembers into adaptive bandwise weights for iterative HSI SR reconstruction, thereby improving cross-sensor generalization.
3) An interpretable optimization-driven network with LSSSC is proposed to integrate spectral–spatial gradient priors and adaptive attention weights, improving spectral preservation while alleviating excessive smoothing near boundaries.

II. Related Works

A. Model- and Data-Driven HSI SR Methods

Early HSI SR methods were mainly model-driven and relied on low-dimensional manifold assumptions, sparse representation, and handcrafted priors [24], [25]. These methods exploit spatial–spectral properties, such as low-rank structure, self-similarity, and total variation, to reconstruct high-resolution HSIs. For example, dictionary learning-based methods use

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paired high- and low-resolution dictionaries with sparse coding [26]. Akgun et al. [27] represented hyperspectral observations as weighted linear combinations of basis image planes. Huang et al. [28] introduced low-rank and group sparse priors for HSI SR with unknown blur. Some methods incorporated nonlocal filtering, 3-D total variation, spectral mixture analysis, and spatial–spectral group sparsity to preserve spectral consistency and spatial details [29], [30]. Although model-driven methods provide good physical interpretability, their ability to reconstruct complex image details is limited.

With the development of deep learning, data-driven HSI SR methods have become the mainstream. Early convolutional neural networks (CNNs) learned end-to-end mappings from low- to high-resolution HSIs. 3DFCNN employs 3-D convolutions to jointly extract spatial and spectral features [13], while the grouped deep recursive residual network (GDRRN) uses a compact grouped recursive structure to model complex nonlinear mappings [14]. MCNet further combines 2-D and 3-D convolutions for mixed spatial–spectral feature extraction [31], and SRDNet processes HSIs in both the spatial and frequency domains [32]. To improve global feature modeling, the hybrid local and nonlocal network (HLNnet) combines local and nonlocal information [15], while Interactformer integrates Transformer and CNN structures. More recently, Mamba- and diffusion-based methods have also been introduced for HSI SR [33], [34]. In addition to task-specific HSI SR, unified HSI restoration has also attracted increasing attention. For example, recent methods utilize prompt learning to address multiple or complex HSI degradation problems within a unified model [18], [35]. Besides the specific HSI SR, recent studies have investigated remote sensing image SR from different perspectives. Kang et al. developed an efficient Swin Transformer for remote sensing image SR [36], while Yao et al. [37] introduced uncertainty estimation into remote sensing SR to improve reconstruction stability. Nevertheless, most data-driven HSI restoration methods still rely on network designs and usually do not explicitly consider cross-sensor generalization in HSI SR.

B. Optimization-Driven HSI SR Methods

Model- and data-driven methods offer complementary advantages. Therefore, the coupling of model- and data-driven methods (i.e., optimization-driven methods) has gained significant attention because it combines physical interpretability with strong learning ability [38], [39]. These methods realize end-to-end training by unfolding iterative optimization algorithms into trainable neural network layers. For example, Liu et al. [12] proposed an efficient spatial–spectral representation unfolding network (EUNet) by integrating prior knowledge (i.e., spectral correlation, degradation model, and structural prior) with deep learning, achieving superior performance in HSI SR. Unlike the regularization module of EUNet, which mainly employs simple 2-D convolutions and attention mechanisms, Wang et al. [40] integrated transformers, 3-D convolutions, and 2-D convolutions into the optimization framework, achieving better performance. However, these methods consume significant computational resources. By integrating the degradation principle of HSI, a CNN- and transformer-embedded unfolding network was designed, which employs a maximum a posteriori-based energy model to seek optimal solutions through alternating optimization [20]. These methods formulate HSI SR as an optimization problem consisting of a data fidelity term and a prior regularization term, which are solved as alternating subproblems to ensure physical consistency and structural integrity, respectively. However, prior learning via neural networks often fails to fully exploit the intrinsic prior of HSIs.

Algorithm 1 Interpretable Cross-Sensor HSI SR Framework

Input: Low-resolution HSI $\mathbf{Y}$, number of endmembers M, and unfolding stages T
Output: Reconstructed HSI $\hat{\mathbf{X}}_T$

Reshape $\mathbf{Y}$ into $\mathbf{Y}_{2D}$
Initialize $\mathbf{X}_0$ by bicubic interpolation
$\hat{\mathbf{X}}_0 \leftarrow \mathbf{X}_0$
Initialize $\mathbf{Z}_0$ by convolution
$\mathbf{E} \leftarrow \mathrm{VCA}(\mathbf{Y}_{2D}, M)$
$\mathbf{w} \leftarrow \mathrm{EDMN}(\mathbf{E})$
for $t = 1, \ldots, T$ **do**
  Fidelity update:
  $\mathbf{Z}_t \leftarrow \mathcal{F}(\mathbf{Z}_{t-1}, \hat{\mathbf{X}}_{t-1}, \mathbf{Y})$
  Prior update:
  $\mathbf{V}_t \leftarrow \mathcal{P}(\hat{\mathbf{X}}_{t-1})$
  Spectral prior update:
  $\mathbf{X}_t \leftarrow \mathcal{S}(\hat{\mathbf{X}}_{t-1}, \mathbf{Z}_t, \mathbf{V}_t)$
  Endmember-aware modulation:
  $\hat{\mathbf{X}}_t \leftarrow \mathbf{X}_t \cdot \mathrm{Expand}(\mathbf{w})$
end for
return $\hat{\mathbf{X}}_T$

III. Methodology

This section details the proposed interpretable cross-sensor HSI SR framework. The overall architecture is illustrated in Fig. 3, while Algorithm 1 summarizes the optimization-driven unfolding process. $\mathcal{F}$, $\mathcal{P}$, and $\mathcal{S}$ denote the fidelity, prior learning, and spectral prior subnetworks, respectively.

A. Endmember-Aware Dynamic Modulation Network

Differences across sensors limit the generalization of existing HSI SR methods to data from unseen sensors. Although endmember signatures may vary across sensors, previous studies have shown that endmember-based representations can provide material-related information for multisensor remote sensing tasks [41]. Motivated by this observation, we propose the EDMN to exploit these transferable material-related priors from endmembers, thereby improving cross-sensor HSI SR generalization. As shown in Fig. 3, the material-related information extracted from endmembers is transformed into adaptive bandwise weights to modulate the iterative HSI SR process.

Given a low-resolution HSI $\mathbf{Y} \in \mathbb{R}^{h \times w \times B}$, where h, w, and B denote the height, width, and number of spectral bands, respectively, we reshape it into $\mathbf{Y}_{2D} \in \mathbb{R}^{hw \times B}$. Based on the

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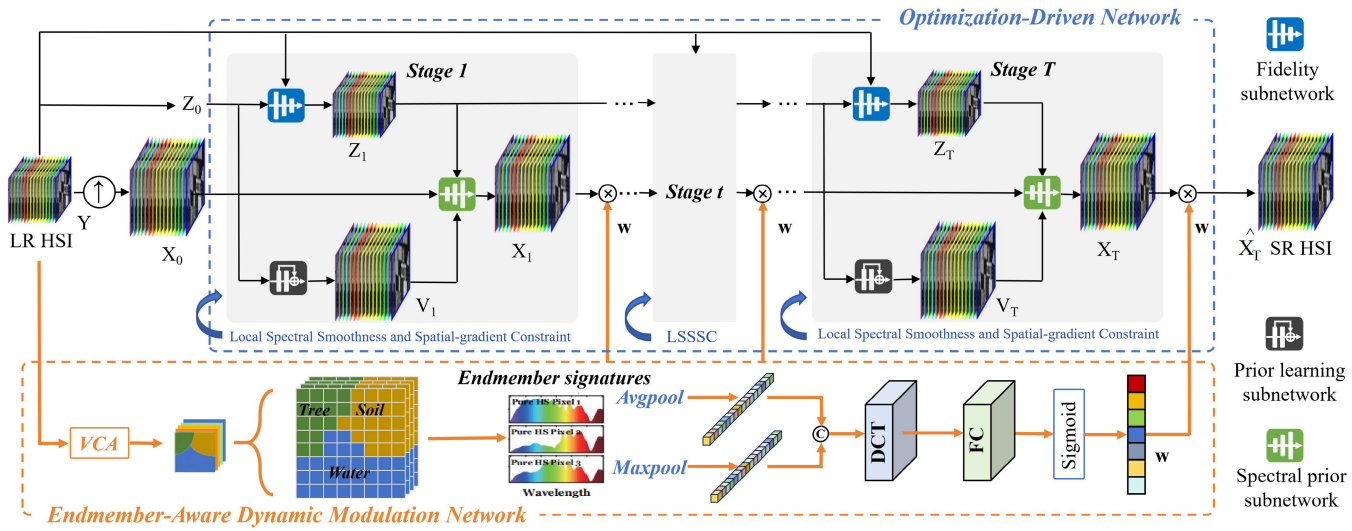


Fig. 3. Architecture of the interpretable cross-sensor HSI SR framework.

linear mixing model, vertex component analysis (VCA) [42] is employed to extract M representative endmember signatures from the input HSI

$$\mathbf{E} = \mathrm{VCA}\left(\mathbf{Y}_{2D}, M\right) \tag{1}$$

where $\mathbf{E} \in \mathbb{R}^{M\times B}$ is the extracted endmember matrix and M is the number of endmembers.

To obtain compact material-related information, max pooling and average pooling are performed along the endmember dimension

$$\mathbf{c}_{\max}(b) = \max_{1\le m\le M} \mathbf{E}_{m,b}, \quad \mathbf{c}_{\mathrm{avg}}(b) = \frac{1}{M}\sum_{m=1}^{M}\mathbf{E}_{m,b} \tag{2}$$

where $b = 1,\ldots,B$ denotes the spectral band index. The pooled statistics are then expressed as follows:

$$\mathbf{C} = \left[\mathbf{c}_{\max}; \mathbf{c}_{\mathrm{avg}}\right] \in \mathbb{R}^{2\times B}. \tag{3}$$

Different from conventional channel attention mechanisms that directly generate weights from the pooled statistics, EDMN first applies a two-point discrete cosine transform (DCT) to the max- and average-pooled statistics at each spectral band

$$\widetilde{\mathbf{C}} = \mathbf{Q}_{\mathrm{DCT}}\mathbf{C}, \qquad \mathbf{Q}_{\mathrm{DCT}} \in \mathbb{R}^{2\times 2} \tag{4}$$

where $\mathbf{Q}_{\mathrm{DCT}}$ denotes the fixed orthonormal basis matrix of the two-point DCT, whose element is defined as follows:

$$[\mathbf{Q}_{\mathrm{DCT}}]_{k,n} = \alpha_k \cos\left[\frac{\pi}{2}\left(n+\frac{1}{2}\right)k\right], \quad k,n\in\{0,1\} \tag{5}$$

where $\alpha_0 = 1/(2)^{1/2}$ and $\alpha_1 = 1$ are normalization factors. The transformed statistics $\widetilde{\mathbf{C}} \in \mathbb{R}^{2\times B}$ contain the two transformed coefficients for each spectral band.

The transformed statistics are then mapped to bandwise modulation weights through a shared linear projection

$$\mathbf{w} = \mathrm{Sigmoid}\left(\widetilde{\mathbf{C}}^{T}\mathbf{W}_m\right) \tag{6}$$

where $\mathbf{W}_m \in \mathbb{R}^{2\times 1}$ denotes the learnable parameters and is shared across all spectral bands. The resulting vector $\mathbf{w}$ provides the modulation weight for each spectral band. The length of $\mathbf{w}$ adaptively follows the number of spectral bands of input HSIs, enabling EDMN to process HSIs with different numbers of bands.

At the tth unfolding stage, the reconstructed HSI is modulated by

$$\hat{\mathbf{X}}_t = \mathbf{X}_t \cdot \mathrm{Expand}\left(\mathbf{w}\right) \tag{7}$$

where $\cdot$ denotes elementwise multiplication and Expand($\mathbf{w}$) broadcasts the bandwise weights along the spatial dimensions to match the shape of $\mathbf{X}_t$. Equation (7) introduces the transferable material-related priors into the iterative reconstruction process rather than acting as an independent operation. Since endmembers represent characteristic spectral signatures of materials, the generated bandwise weights provide spectral guidance for each unfolding stage. The modulated result is subsequently fed into the next unfolding stage, thereby coupling the endmember-aware dynamic modulation with the iterative HSI SR reconstruction.

B. Optimization-Driven Network With LSSSC

The observed low-resolution HSI is denoted by $\mathbf{Y} \in \mathbb{R}^{h\times w\times B}$, and the corresponding high-resolution HSI is denoted by $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where $H = hs$, $W = ws$, and s is the scale factor. After reshaping the 3-D HSI cubes into 2-D matrices, we have $\mathbf{Y} \in \mathbb{R}^{n_l\times B}$ and $\mathbf{X} \in \mathbb{R}^{n_h\times B}$, where $n_l = hw$ and $n_h = HW$. The degradation model is written as follows:

$$\mathbf{Y} = \mathbf{HDX} + \mathbf{N} \tag{8}$$

where $\mathbf{D} \in \mathbb{R}^{n_l\times n_h}$ denotes the spatial downsampling operator from the high-resolution images to the low-resolution images via bicubic downsampling, $\mathbf{H} \in \mathbb{R}^{n_l\times n_l}$ denotes the blur operator on the low-resolution images using a Gaussian blur kernel, and $\mathbf{N} \in \mathbb{R}^{n_l\times B}$ denotes additive independent noise. $\mathbf{H}$ and $\mathbf{D}$ are fixed degradation operators.

To recover $\mathbf{X}$ from $\mathbf{Y}$, an optimization model with data fidelity and prior regularization is adopted

$$\mathbf{X}^{*} = \arg\min_{\mathbf{X}} \frac{1}{2\sigma^2}\|\mathbf{Y} - \mathbf{HDX}\|_F^2 + \lambda\Phi\left(\mathbf{X}\right) \tag{9}$$

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where σ denotes the noise standard deviation, $\Phi(\mathbf{X})$ denotes the learned HSI prior, and λ is the regularization parameter. $\|\cdot\|_F$ denotes the Frobenius norm.

HSIs generally exhibit smooth spectral signatures within homogeneous regions, whereas pixels near boundaries are more likely to contain mixtures of different materials. Uniform spectral smoothing may suppress material transitions and cause spectral distortion near boundaries. Therefore, LSSSC is incorporated into the optimization model

$$\mathbf{X}^* = \arg\min_{\mathbf{X}} \frac{1}{2\sigma^2}\|\mathbf{Y}-\mathbf{HDX}\|_F^2 + \lambda_1\Phi(\mathbf{X}) + \lambda_2\left(\|\nabla_s\mathbf{X}\|_F^2 + \|\nabla_h\mathbf{X}\|_F^2 + \|\nabla_w\mathbf{X}\|_F^2\right) \tag{10}$$

where ∇_s, ∇_h, and ∇_w denote the spectral, vertical spatial, and horizontal spatial gradient operators, respectively. Let $\mathcal{R}_{\mathrm{LSSSC}}(\mathbf{X})$ denote the sum of the three gradient regularization terms. Its gradient with respect to $\mathbf{X}$ is

$$\nabla\mathcal{R}_{\mathrm{LSSSC}}(\mathbf{X}) = 2\left(\nabla_s^T\nabla_s\mathbf{X} + \nabla_h^T\nabla_h\mathbf{X} + \nabla_w^T\nabla_w\mathbf{X}\right). \tag{11}$$

In homogeneous regions, the spatial-gradient responses $|\nabla_h\mathbf{X}|$ and $|\nabla_w\mathbf{X}|$ are usually small, indicating local consistency. In these regions, spectral smoothness helps maintain spectral continuity. In contrast, boundary regions often correspond to material transitions or mixed pixels, where imposing the same spectral smoothing strength may cause spectral distortion. LSSSC provides explicit spectral–spatial gradient priors, while the spectral prior subnetwork uses adaptive attention weights to adjust the contributions of different gradient branches. Therefore, LSSSC and adaptive attention weights jointly promote spectral smoothness in homogeneous regions while mitigating excessive smoothing near boundaries.

To solve the above optimization problem, two auxiliary variables $\mathbf{Z}$ and $\mathbf{V}$ are introduced

$$\mathbf{Z} = \mathbf{DX}, \quad \mathbf{V} = \mathbf{X}. \tag{12}$$

Using variable splitting techniques [43], the optimization problem is approximated as follows:

$$L(\mathbf{X},\mathbf{Z},\mathbf{V}) = \frac{1}{2\sigma^2}\|\mathbf{Y}-\mathbf{HZ}\|_F^2 + \lambda_1\Phi(\mathbf{V}) + \lambda_2\left(\|\nabla_s\mathbf{X}\|_F^2 + \|\nabla_h\mathbf{X}\|_F^2 + \|\nabla_w\mathbf{X}\|_F^2\right) + \frac{\mu_1}{2}\|\mathbf{Z}-\mathbf{DX}\|_F^2 + \frac{\mu_2}{2}\|\mathbf{V}-\mathbf{X}\|_F^2 \tag{13}$$

where μ_1 and μ_2 are penalty parameters. The corresponding subproblems are

$$\begin{cases} \mathbf{Z} &= \arg\min_{\mathbf{Z}} \frac{1}{2\sigma^2}\|\mathbf{Y}-\mathbf{HZ}\|_F^2 + \frac{\mu_1}{2}\|\mathbf{Z}-\mathbf{DX}\|_F^2 \\ \mathbf{V} &= \arg\min_{\mathbf{V}} \frac{\mu_2}{2}\|\mathbf{V}-\mathbf{X}\|_F^2 + \lambda_1\Phi(\mathbf{V}) \\ \mathbf{X} &= \arg\min_{\mathbf{X}} \frac{\mu_1}{2}\|\mathbf{Z}-\mathbf{DX}\|_F^2 + \frac{\mu_2}{2}\|\mathbf{V}-\mathbf{X}\|_F^2 \\ &\quad +\lambda_2\left(\|\nabla_s\mathbf{X}\|_F^2 + \|\nabla_h\mathbf{X}\|_F^2 + \|\nabla_w\mathbf{X}\|_F^2\right) \end{cases} \tag{14}$$

where the $\mathbf{Z}$-subproblem enforces data consistency, the $\mathbf{V}$-subproblem learns HSI priors through a neural network, and the $\mathbf{X}$-subproblem combines these two constraints with LSSSC to preserve spectral fidelity.

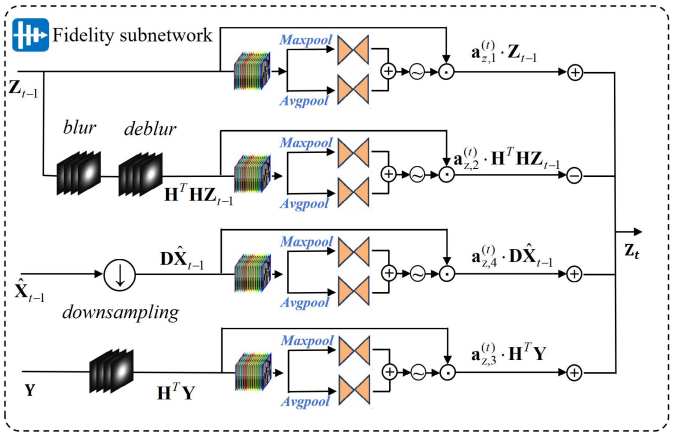


Fig. 4. Architecture of fidelity subnetwork.

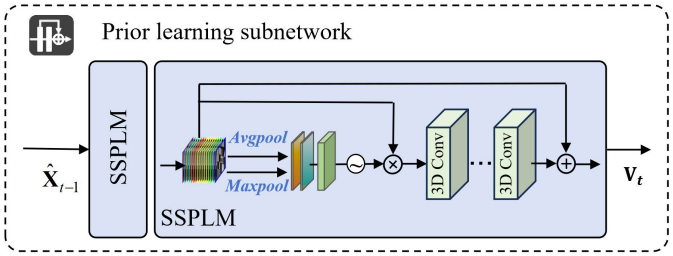


Fig. 5. Architecture of prior learning subnetwork.

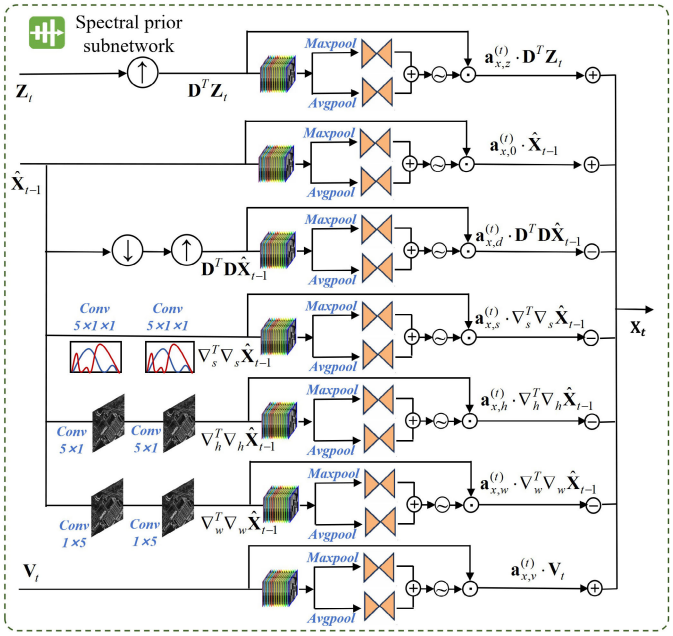


Fig. 6. Architecture of spectral prior subnetwork.

C. Fidelity, Prior Learning, and Spectral Prior Subnetworks

Based on the above iterative formulation, three subnetworks are designed to approximate the solutions of the $\mathbf{Z}$-, $\mathbf{V}$-, and $\mathbf{X}$-subproblems, namely, the fidelity subnetwork, the prior learning subnetwork, and the spectral prior subnetwork, as shown in Figs. 4–6. Instead of solving each subproblem exactly, each unfolding stage performs a learnable update to approximate the solution of the corresponding subproblem. This design maintains the interpretability of iterative optimization while combining explicit spectral–spatial gradient priors.

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As shown in Fig. 4, the fidelity subnetwork addresses the **Z**-subproblem and enforces data consistency between the observed low-resolution HSI and the degraded reconstruction. For the **Z**-subproblem, we define

$$f_Z(\mathbf{Z}) = \frac{1}{2}\|\mathbf{Y} - \mathbf{HZ}\|_F^2 + \frac{\mu_1\sigma^2}{2}\left\|\mathbf{Z} - \mathbf{D}\hat{\mathbf{X}}_{t-1}\right\|_F^2. \tag{15}$$

Following [12], instead of solving the **Z**-subproblem exactly, the update of $\mathbf{Z}_t$ is approximated at each unfolding stage via a single gradient descent step from $\mathbf{Z}_{t-1}$

$$\mathbf{Z}_t = \mathbf{Z}_{t-1} - \tau\nabla f_Z(\mathbf{Z}_{t-1}) \tag{16}$$

where τ is the gradient step size. The optimization-derived update is

$$\begin{aligned}\mathbf{Z}_t &= \mathbf{Z}_{t-1} - \tau\left[\mathbf{H}^T(\mathbf{HZ}_{t-1} - \mathbf{Y}) + \mu_1\sigma^2\left(\mathbf{Z}_{t-1} - \mathbf{D}\hat{\mathbf{X}}_{t-1}\right)\right] \\ &= \left(1 - \tau\mu_1\sigma^2\right)\mathbf{Z}_{t-1} - \tau\mathbf{H}^T\mathbf{HZ}_{t-1} \\ &\quad + \tau\mathbf{H}^T\mathbf{Y} + \tau\mu_1\sigma^2\mathbf{D}\hat{\mathbf{X}}_{t-1}\end{aligned} \tag{17}$$

where t is the iteration index. In the implemented fidelity subnetwork, the fixed coefficients derived from the optimization model are replaced by adaptive attention weights to adjust the contributions of different branches. Accordingly, the implemented update is reformulated as follows:

$$\begin{aligned}\mathbf{Z}_t = \mathbf{a}_{z,1}^{(t)} \cdot \mathbf{Z}_{t-1} &- \mathbf{a}_{z,2}^{(t)} \cdot \mathbf{H}^T\mathbf{HZ}_{t-1} + \mathbf{a}_{z,3}^{(t)} \cdot \mathbf{H}^T\mathbf{Y} \\ &+ \mathbf{a}_{z,4}^{(t)} \cdot \mathbf{D}\hat{\mathbf{X}}_{t-1}\end{aligned} \tag{18}$$

where $\cdot$ denotes elementwise multiplication. The generated adaptive weights are broadcast to their corresponding branch features for elementwise modulation. The adaptive weights $\mathbf{a}_{z,i}^{(t)}$ are generated by the fidelity subnetwork from the corresponding channel attention network

$$\mathbf{a}_{z,i}^{(t)} = \mathcal{A}_{z,i}^{(t)}\left(\mathbf{F}_{z,i}^{(t)}; \mathbf{\Theta}_{z,i}\right), \quad i = 1, \ldots, 4 \tag{19}$$

where $\mathcal{A}_{z,i}^{(t)}(\cdot)$ denotes the attention-based weight generation function, $\mathbf{F}_{z,i}^{(t)}$ denotes the feature of the ith branch, and $\mathbf{\Theta}_{z,i}$ denotes the learnable network parameters.

In addition, the prior learning subnetwork addresses the **V**-subproblem by learning spatial–spectral correlations in HSIs. Using a proximal operator formulation

$$\mathbf{V}_t = \text{Prox}\left(\hat{\mathbf{X}}_{t-1}\right) = \arg\min_{\mathbf{V}} \frac{\mu_2}{2}\left\|\mathbf{V} - \hat{\mathbf{X}}_{t-1}\right\|_F^2 + \lambda_1\Phi(\mathbf{V}). \tag{20}$$

In the proposed network, this proximal mapping is approximated by the prior learning subnetwork

$$\mathbf{V}_t = \mathcal{P}\left(\hat{\mathbf{X}}_{t-1}; \mathbf{\Theta}_p\right) \tag{21}$$

where $\mathcal{P}(\cdot)$ denotes the spectral–spatial prior learning subnetwork. As shown in Fig. 5, this subnetwork consists of spectral–spatial prior learning modules (SSPLMs) that integrate 3-D convolutions and spatial attention mechanisms. It learns implicit HSI priors from the training data.

After obtaining $\mathbf{Z}_t$ and $\mathbf{V}_t$ from the fidelity subnetwork and the prior learning subnetwork, respectively, the spectral prior subnetwork addresses the **X**-subproblem by combining learned HSI priors with explicit spectral–spatial gradient priors, as shown in Fig. 6. A gradient descent update is performed as follows:

$$\begin{aligned}\mathbf{X}_t = \hat{\mathbf{X}}_{t-1} - \varepsilon\Big[&\mu_1\mathbf{D}^T\left(\mathbf{D}\hat{\mathbf{X}}_{t-1} - \mathbf{Z}_t\right) + \mu_2\left(\hat{\mathbf{X}}_{t-1} - \mathbf{V}_t\right) \\ &+ 2\lambda_2\left(\nabla_s^T\nabla_s\hat{\mathbf{X}}_{t-1} + \nabla_h^T\nabla_h\hat{\mathbf{X}}_{t-1} + \nabla_w^T\nabla_w\hat{\mathbf{X}}_{t-1}\right)\Big]\end{aligned} \tag{22}$$

where ε is the gradient step size. This formulation shows how LSSSC is incorporated into the gradient update through the three terms $\nabla_s^T\nabla_s\hat{\mathbf{X}}_{t-1}$, $\nabla_h^T\nabla_h\hat{\mathbf{X}}_{t-1}$, and $\nabla_w^T\nabla_w\hat{\mathbf{X}}_{t-1}$.

In the implemented spectral prior subnetwork, the fixed coefficients derived from the optimization model are replaced by adaptive attention weights

$$\begin{aligned}\mathbf{X}_t = \mathbf{a}_{x,0}^{(t)} \cdot \hat{\mathbf{X}}_{t-1} &- \mathbf{a}_{x,d}^{(t)} \cdot \mathbf{D}^T\mathbf{D}\hat{\mathbf{X}}_{t-1} - \mathbf{a}_{x,s}^{(t)} \cdot \nabla_s^T\nabla_s\hat{\mathbf{X}}_{t-1} \\ &- \mathbf{a}_{x,h}^{(t)} \cdot \nabla_h^T\nabla_h\hat{\mathbf{X}}_{t-1} - \mathbf{a}_{x,w}^{(t)} \cdot \nabla_w^T\nabla_w\hat{\mathbf{X}}_{t-1} \\ &+ \mathbf{a}_{x,z}^{(t)} \cdot \mathbf{D}^T\mathbf{Z}_t + \mathbf{a}_{x,v}^{(t)} \cdot \mathbf{V}_t.\end{aligned} \tag{23}$$

The adaptive weights are generated as follows:

$$\mathbf{a}_{x,j}^{(t)} = \mathcal{A}_{x,j}^{(t)}\left(\mathbf{F}_{x,j}^{(t)}; \mathbf{\Theta}_{x,j}\right), \quad j \in \{0, d, s, h, w, z, v\} \tag{24}$$

where $\mathbf{F}_{x,j}^{(t)}$ denotes the feature of the corresponding branch. The spectral-gradient branch helps maintain spectral continuity, while the spatial-gradient branches capture boundary variations. Spatial-gradient responses are generally small in homogeneous regions but become large near boundaries or mixed pixels. The attention mechanism in each branch adaptively adjusts its contribution according to the corresponding feature. Therefore, LSSSC and adaptive attention weights jointly promote spectral smoothness in homogeneous regions while mitigating excessive smoothing near boundaries.

D. Loss Function

In contrast to conventional optimization-driven methods that supervise only the final output, a deep supervision strategy is employed to constrain the modulated output of each unfolding stage. The total loss is defined as follows:

$$L_{\text{total}} = \sum_{t=1}^{T}\left(L_1^{(t)} + L_{\text{SAM}}^{(t)}\right) \tag{25}$$

where T is the number of unfolding stages, $L_1^{(t)}$ denotes the L_1 loss at stage t, and $L_{\text{SAM}}^{(t)}$ denotes the spectral angle mapper (SAM) loss at stage t. This deep supervision strategy progressively improves spatial details and spectral fidelity throughout the unfolding process.

At each stage t, the L_1 loss measures the absolute difference between the modulated reconstructed HSI and the ground truth

$$L_1^{(t)} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{X}_g^i - \hat{\mathbf{X}}_t^i\right\|_1 \tag{26}$$

where $\hat{\mathbf{X}}_t^i$ and $\mathbf{X}_g^i$ denote the reconstructed and ground truth spectral vectors at the ith pixel, respectively, and N is the total number of pixels. This loss maintains pixelwise intensity fidelity and helps recover spatial details.

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The $L_{\mathrm{SAM}}^{(t)}$ loss measures spectral similarity by treating the spectrum of each pixel as a B-dimensional vector

$$L_{\mathrm{SAM}}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} \arccos\left(\frac{\mathbf{X}_g^i \cdot \hat{\mathbf{X}}_t^i}{\left\|\mathbf{X}_g^i\right\|_2 \left\|\hat{\mathbf{X}}_t^i\right\|_2}\right). \tag{27}$$

IV. Experiments

A. Datasets

Three datasets are used in the experiments, i.e., Chikusei [44], Pavia Center [45], and Houston[1] datasets. The Chikusei dataset is employed for model training, while all three datasets are utilized for testing. Due to differences in spectral ranges and numbers of bands across the three datasets, the overall performance and cross-sensor generalization of the proposed method can be verified.

1) Chikusei Dataset: The Chikusei dataset, an airborne hyperspectral dataset captured by the Headwall Hyperspec-VNIR-C imaging sensor, covers agricultural and urban areas in Chikusei, Ibaraki, Japan, and contains 128 bands. An intermediate region (2318×2136 pixels) is selected from Chikusei, with 80% designated as the training region. Specifically, eight nonoverlapping $256 \times 256 \times 128$ pixel patches are extracted for testing data, while the remaining area is cropped into $64 \times 64 \times 128$-pixel training patches.

2) Pavia Center Dataset: The Pavia Center dataset, an airborne hyperspectral dataset acquired by the Reflective Optics System Imaging Spectrometer, covers the urban center of Pavia, Italy. With 102 spectral bands in the 430–860-nm range, it provides detailed spectral information for urban land cover classification. A 1096×715 pixel region of interest is selected, from which eight nonoverlapping $256 \times 256 \times 102$ pixel patches are cropped as testing data.

3) Houston Dataset: The Houston 2018 dataset, acquired by the HSI Analysis Group and the Airborne Laser Mapping Center at the University of Houston, covers the university campus and surrounding areas. It has a spatial size of 601×2384 pixels, with 48 spectral bands. Eight $512 \times 512 \times 48$ pixel patches are cropped for testing.

For simulation experiments, we follow Wald's protocol [46] to generate low- and high-resolution image pairs. Specifically, the original images are downsampled with scale factors of 2, 4, and 8. In the evaluation of no-reference visual quality, the SR with a scaling factor of 4 is applied to eight nondegraded images selected from the Chikusei, Pavia Center, and Houston datasets, respectively.

B. Comparison Algorithms and Evaluation Metrics

The proposed method is compared with twelve HSI SR methods, including bicubic interpolation and 11 learning-based SR methods: 3DFCNN [13], GDRRN [14], HLNnet [15], MCNet [31], EUNet [12], Interactformer [10], SNLSR [16], and SRDNet [32], state-of-the-art SR method of natural images (CATANet) [47], MPHSIR [18], and WESSP [48]. For a fair comparison, all methods are trained using the same training data and degradation settings under the same experimental environment.

[1]https://hyperspectral.ee.uh.edu/?page_id=1075

In experiments on simulated data, five widely used quantitative metrics are employed, including peak signal-to-noise ratio (PSNR), SAM, erreur relative globale adimensionnelle de synthése (ERGAS), root mean square error (RMSE), and structural similarity index (SSIM) [49], [50], [51]. In addition, for experiments on real data, where ground truth is unavailable, the natural image quality evaluator (NIQE) is utilized as a no-reference metric to assess the perceptual quality of the SR HSIs [31].

C. Implementation Details

The proposed network is implemented on an NVIDIA RTX 4090 GPU using the PyTorch framework, with scale factors of 2, 4, and 8. During training, the batch size is set to 16, and the Adam optimizer [52] is used with an initial learning rate of $1e^{-4}$. In the experiments, the number of iterations T is set to 5. In the EDMN, the number of endmembers is set to 20. In the prior learning subnetwork, the number of SSPLMs is 2, and the number of 3-D convolutions in SSPLM is 3. The spectral prior subnetwork employs fixed gradient filters with kernel sizes of $5 \times 1 \times 1$, 5×1, and 1×5 to compute spectral, vertical spatial, and horizontal spatial gradients, respectively. To process HSIs with different numbers of spectral bands, the spectral dimension is treated as the depth dimension rather than the input channel dimension in 3-D convolutions. Therefore, the 3-D convolution kernels can process HSIs with different spectral dimensions. For 2-D convolutions, each spectral band is reshaped into a single-band image and processed using shared convolutional weights. The outputs are then reshaped back into the HSI cube. The comparison methods are also adapted in this manner to conduct cross-sensor experiments.

D. Experiments on the Simulated Data

For the HSI SR experiments, same-sensor evaluations are conducted on the Chikusei dataset. In addition, to assess cross-sensor generalization, the model is trained on Chikusei and directly evaluated on the Pavia Center and Houston datasets. Bold and underline indicate the best and second-best results, respectively.

1) Performance of Same-Sensor HSI SR on Chikusei Dataset: In this experiment, training and testing are performed on the same sensor dataset. Table I compares the proposed method with several representative methods on the Chikusei dataset. Across scale factors of $\times 2$, $\times 4$, and $\times 8$, the proposed method achieves the best or second-best results on all five metrics on the Chikusei dataset. Bicubic interpolation obtains the lowest scores because it relies on simple pixel interpolation and cannot recover fine spatial details. Among deep learning methods, EUNet combines iterative optimization with learned HSI priors, while the proposed method further incorporates explicit spectral–spatial gradient priors and adaptive attention weights. By combining knowledge- and data-driven modeling, the proposed method produces higher-quality HSIs. For example, at $4\times$ SR, it achieves a PSNR of 31.4695 dB, exceeding the second-best WESSP (31.3020 dB) by 0.1675 dB. At $2\times$, the ERGAS and RMSE values are reduced to 2.5354 and

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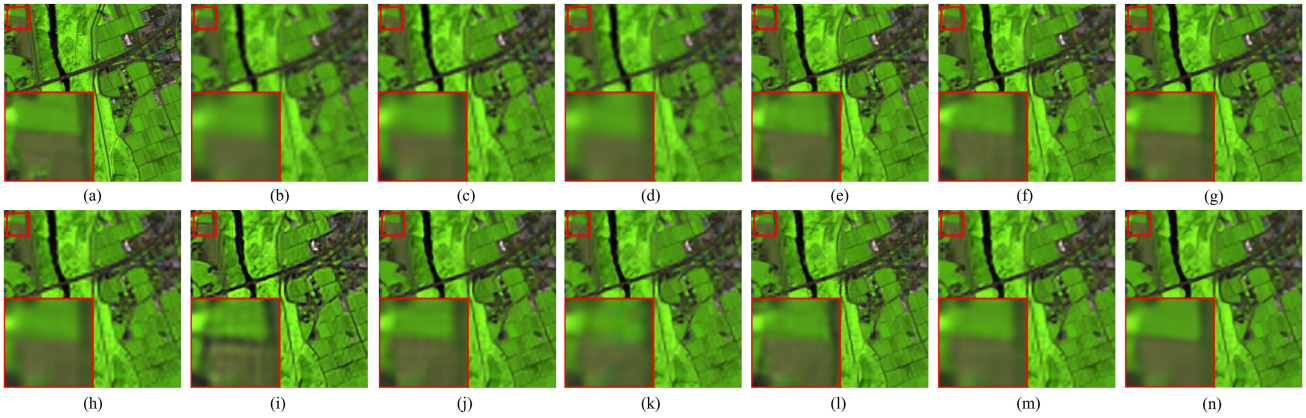


Fig. 7. Reconstructed HSI on the Chikusei dataset with spectral bands 100-30-12 as R-G-B with the scale factor $\times 4$. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

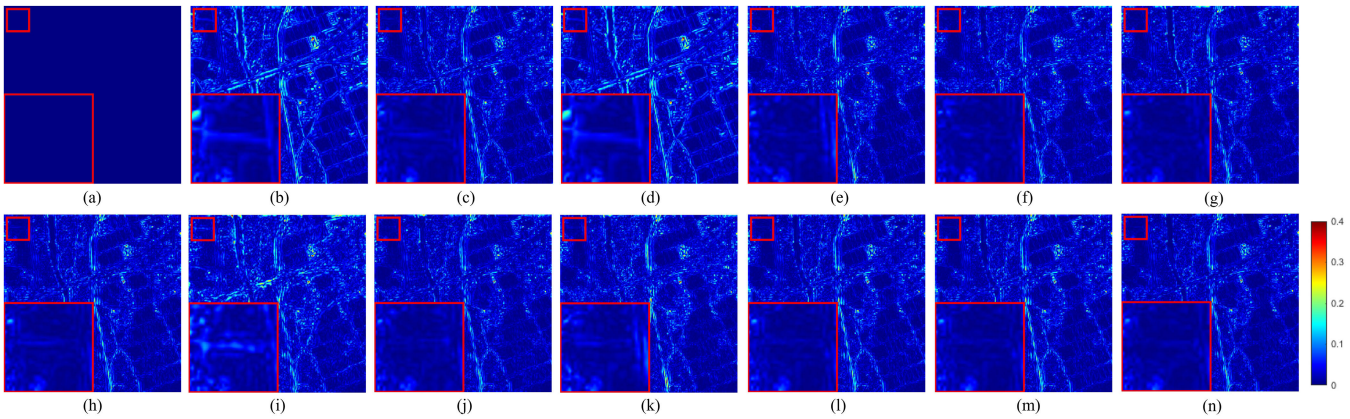


Fig. 8. Ground truth and error maps of the reconstructed HSIs on the Chikusei dataset at a scale factor of $\times 4$. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

0.0172, respectively, indicating lower reconstruction errors. The qualitative results in Figs. 7 and 8 further support the quantitative results. The proposed method reconstructs HSIs with sharper edges and clearer spatial details, consistent with its high SSIM values. The error maps in Fig. 8 also show that the proposed method produces smaller reconstruction errors, demonstrating its better visual quality on the Chikusei dataset.

2) Performance of Cross-Sensor HSI SR on Pavia Center and Houston Datasets: To evaluate cross-sensor HSI SR performance, the models trained on the Chikusei dataset are directly tested on the Pavia Center and Houston datasets. The quantitative results in Table I demonstrate the performance of the proposed method. The proposed method achieves better performance on most metrics across the two cross-sensor datasets. Bicubic interpolation does not depend on training data and is, therefore, unaffected by sensor-specific training. However, it cannot recover fine spatial and spectral details. Data-driven methods, such as 3DFCNN and GDRRN, show limited performance when directly applied to unseen sensor data. EUNet, as an optimization-driven method, provides better interpretability but still struggles to reduce the differences across sensors. In contrast, the proposed method exploits transferable material-related priors from endmembers to improve cross-sensor generalization. Specifically, on the Pavia Center dataset at $4\times$ SR, the proposed method outperforms SRDNet by 0.0815 dB in PSNR, 0.5396 in SAM, 0.0108 in ERGAS, 0.0005 in RMSE, and 0.0003 in SSIM. It still maintains good performance on the Houston dataset. These results demonstrate the effectiveness of the proposed method for cross-sensor HSI SR. The qualitative results further support the quantitative analysis. As shown in Figs. 9 and 10, the proposed method reconstructs HSIs with sharper edges and clearer structures on the Pavia Center dataset, while comparison methods produce blurrier results. Similar observations can be made on the Houston dataset in Figs. 11 and 12. The error maps show smaller reconstruction errors than WESSP. Moreover, the proposed method restores fine details more accurately, whereas other methods tend to lose spatial details or introduce artifacts, further demonstrating its performance in cross-sensor HSI SR.

To compare the proposed method with other methods in preserving the spectral correlation of HSIs, we show the reconstruction results along the spectral dimension via spectral curves, as illustrated in Fig. 13. Pixels are randomly selected from test images to compare the spectral similarity between the ground truth and SR images across three datasets. Spectral curves precisely reflect the values of different spectral bands in HSIs. Experimental results show that the reconstructed images from the proposed method most closely match the ground truth. Notably, whether tested on the same-sensor dataset (e.g., Chikusei) or cross-sensor scenarios (e.g., Pavia Center and Houston), the method effectively preserves spectral correlation. This may be because the spectral priors of HSIs

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TABLE I

QUANTITATIVE COMPARISON ON CHIKUSEI, PAVIA CENTER, AND HOUSTON DATASETS AT SCALE FACTORS OF 2, 4, AND 8. THE MODEL IS TRAINED ONLY ON THE CHIKUSEI DATASET AND DIRECTLY TESTED ON THE THREE DATASETS, RESPECTIVELY

		Chikusei Dataset					Pavia Centre Dataset (Cross-sensor)					Houston Dataset (Cross-sensor)				
		PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
Bicubic	×2	35.1562	2.0526	3.9425	0.0273	0.9906	31.4933	5.1608	4.9770	0.0412	0.9622	39.2648	1.5558	1.8574	0.0137	0.9928
$3DFCNN_{2017}$	×2	37.6748	1.7442	3.1235	0.0204	0.9948	33.5436	5.0529	4.0335	0.0325	0.9780	41.6117	1.4542	1.7366	0.0105	0.9958
$GDRRN_{2018}$	×2	36.1495	1.9569	3.5469	0.0233	0.9946	32.4793	5.1050	4.3835	0.0382	0.9721	41.5166	1.4321	1.7521	0.0118	0.9947
$HLNnet_{2021}$	×2	38.6523	1.5757	2.6562	0.0184	0.9957	34.1988	4.6206	3.6805	0.0302	0.9812	42.3642	1.2565	1.4483	0.0096	0.9965
$MCNet_{2020}$	×2	38.7030	1.5129	2.6986	0.0182	0.9959	34.3151	4.5379	3.6548	0.0297	0.9817	42.1632	1.3497	1.5985	0.0098	0.9963
$EUNet_{2023}$	×2	38.7911	1.4853	2.6316	0.0180	0.9959	34.3165	4.4304	3.6591	0.0297	0.9816	42.5104	1.2175	**1.3135**	0.0094	0.9966
$Interactformer_{2022}$	×2	38.7679	**1.3255**	2.5822	0.0180	0.9959	34.4547	4.3087	3.6229	0.0295	0.9815	42.3560	1.1985	1.3554	0.0096	0.9964
$SNLSR_{2024}$	×2	38.8871	1.5221	2.5827	0.0200	0.9956	34.2846	4.2966	3.6915	0.0299	0.9818	42.0116	1.2231	1.3618	0.0096	0.9962
$SRDNet_{2024}$	×2	38.9580	1.5030	2.6560	0.0179	0.9960	34.3225	4.5159	3.6549	0.0297	0.9817	42.4709	1.2574	1.4543	0.0095	0.9966
$CATANet_{2025}$	×2	38.2534	1.7057	2.8150	0.0192	0.9954	34.0657	4.8341	3.7553	0.0306	0.9806	41.8668	1.4817	1.5171	0.0102	0.9960
$MPHSIR_{2025}$	×2	38.9801	1.3821	2.5643	0.0176	0.9961	34.3850	4.3121	3.6353	0.0296	0.9818	42.6000	1.1906	1.3415	0.0094	0.9966
$WESSP_{2026}$	×2	39.0573	1.3652	2.5533	0.0174	0.9961	34.4121	4.2982	3.6255	0.0296	0.9819	42.6438	1.1852	1.3350	0.0093	0.9966
Ours	×2	**39.1952**	1.3435	**2.5354**	**0.0172**	**0.9963**	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**	**42.7569**	**1.1786**	1.3271	**0.0092**	**0.9967**
Bicubic	×4	28.7021	4.2947	8.0813	0.0574	0.9575	26.0185	7.5196	8.9498	0.0771	0.8562	31.2685	3.6291	4.5971	0.0344	0.9559
$3DFCNN_{2017}$	×4	30.5310	3.6441	6.7003	0.0465	0.9727	27.6771	7.3248	7.5676	0.0636	0.9090	33.8796	3.5997	4.2389	0.0254	0.9757
$GDRRN_{2018}$	×4	29.7006	4.0956	7.0823	0.0504	0.9675	27.0172	7.3259	8.3511	0.0671	0.8962	32.2674	3.3299	4.4977	0.0294	0.9659
$HLNnet_{2021}$	×4	30.8609	3.5962	6.0310	0.0450	0.9746	27.7267	7.4939	7.3433	0.0633	0.9136	34.5497	3.0100	3.4635	0.0236	0.9795
$MCNet_{2020}$	×4	30.8398	3.5408	5.9764	0.0450	0.9747	27.8173	7.4501	7.2672	0.0627	0.9151	34.5718	3.1289	3.3873	0.0235	0.9793
$EUNet_{2023}$	×4	31.1068	3.3002	5.9503	0.0436	0.9759	28.0554	6.9518	7.1223	0.0609	0.9189	34.5020	2.8513	**3.1941**	0.0237	0.9793
$Interactformer_{2022}$	×4	30.9313	3.2397	6.1263	0.0444	0.9754	28.0837	6.8215	7.1823	0.0608	0.9179	34.8685	2.6364	3.2446	0.0227	0.9801
$SNLSR_{2024}$	×4	30.6208	3.3188	6.1329	0.0445	0.9745	27.9523	6.7725	8.1940	0.0634	0.9094	33.9947	3.0978	3.2442	0.0232	0.9806
$SRDNet_{2024}$	×4	31.2561	**2.9466**	6.1266	0.0434	0.9741	28.1451	6.9742	7.0618	0.0603	0.9188	34.8798	2.6428	3.2336	0.0223	0.9809
$CATANet_{2025}$	×4	30.6566	4.2854	6.4755	0.0459	0.9731	27.5132	7.4518	7.5957	0.0649	0.9046	33.2457	3.3261	3.7246	0.0274	0.9709
$MPHSIR_{2025}$	×4	31.2215	3.0502	5.9157	0.0425	0.9765	28.1850	6.6520	7.1020	0.0605	0.9185	35.0513	2.6372	3.2531	0.0221	0.9808
$WESSP_{2026}$	×4	31.3020	3.1537	5.8904	0.0422	0.9770	28.1920	6.5510	7.0850	0.0604	0.9187	35.1012	2.6251	3.2417	0.0221	0.9810
Ours	×4	**31.4695**	2.9574	**5.8720**	**0.0419**	**0.9775**	**28.2266**	**6.4346**	**7.0510**	**0.0598**	**0.9191**	**35.1758**	**2.6143**	3.3316	**0.0219**	**0.9811**
Bicubic	×8	25.3452	6.5942	11.7373	0.0846	0.9075	22.8606	10.0317	12.7018	0.1106	0.6884	26.3202	6.6561	8.1347	0.0609	0.8642
$3DFCNN_{2017}$	×8	27.0806	5.4052	9.6830	0.0692	0.9380	24.5765	9.3011	10.5100	0.0908	0.7960	29.2242	5.5674	6.0659	0.0435	0.9280
$GDRRN_{2018}$	×8	26.3979	6.0532	10.6844	0.0801	0.9292	23.9207	9.3142	11.6201	0.1098	0.7748	27.4560	5.5841	7.0092	0.0499	0.8987
$HLNnet_{2021}$	×8	26.5299	6.0214	9.7230	0.0738	0.9302	23.6983	9.7087	11.4104	0.1006	0.7602	28.5538	6.3532	7.3117	0.0470	0.9191
$MCNet_{2020}$	×8	27.1237	5.2676	9.5367	0.0689	0.9382	24.5726	8.9152	10.4801	0.0909	0.7948	29.2779	5.0374	5.9473	0.0433	0.9285
$EUNet_{2023}$	×8	27.3387	5.1192	9.3834	0.0672	0.9413	24.6353	8.7293	10.4070	0.0903	0.8008	29.5852	4.7017	5.6305	0.0417	0.9337
$Interactformer_{2022}$	×8	27.3024	5.0607	9.3194	0.0675	0.9410	24.7193	8.7176	10.3115	0.0896	0.8025	29.7249	4.6941	**5.5781**	0.0411	0.9352
$SNLSR_{2024}$	×8	**27.4626**	5.3018	9.3915	0.0671	0.9409	24.6657	9.0467	10.6312	0.0902	0.7998	29.3064	4.6077	5.8181	0.0422	0.9356
$SRDNet_{2024}$	×8	27.2160	5.2114	9.4816	**0.0663**	0.9408	24.6337	8.9878	**10.2807**	0.0907	0.8007	28.8642	5.0492	5.9314	0.0409	0.9313
$CATANet_{2025}$	×8	27.2819	5.2577	9.4516	0.0677	**0.9417**	24.6663	8.8607	10.3716	0.0900	0.8024	29.4296	5.2878	5.7679	0.0425	0.9305
$MPHSIR_{2025}$	×8	27.2805	5.0854	9.2871	0.0671	0.9412	24.7527	8.6822	10.3352	0.0894	0.8022	29.7503	4.6517	5.6257	0.0408	0.9358
$WESSP_{2026}$	×8	27.3024	5.0531	9.2718	0.0669	0.9414	24.7613	8.6651	10.3215	0.0891	0.8024	29.8012	4.5803	5.5916	0.0405	0.9360
Ours	×8	27.4319	**5.0124**	**9.2412**	0.0668	0.9416	**24.7923**	**8.6412**	10.3011	**0.0886**	**0.8027**	**29.8414**	**4.5213**	5.5782	**0.0402**	**0.9362**

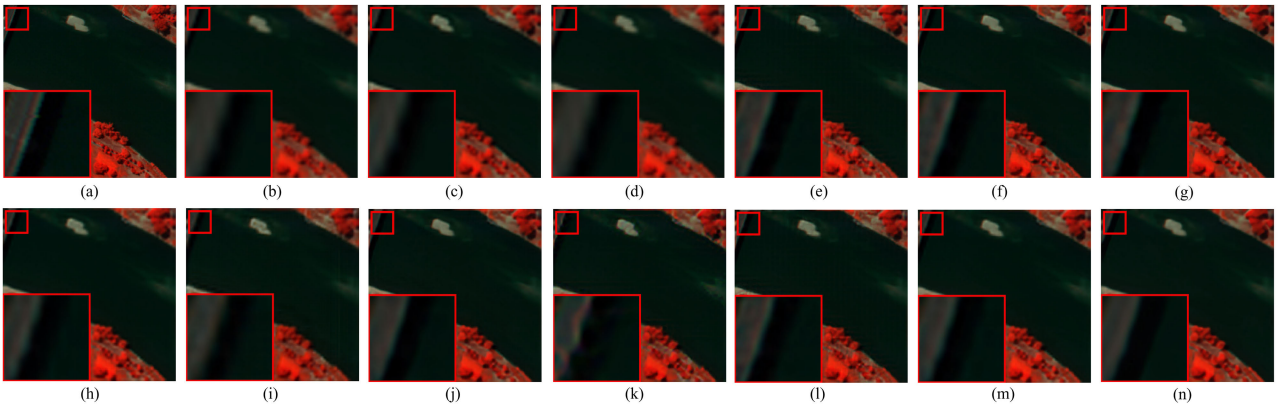


Fig. 9. Reconstructed HSI on the Pavia Center dataset with spectral bands 70-100-36 as R-G-B with the scale factor ×4. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

are effectively leveraged by integrating the designed LSSSC, ensuring accurate preservation of spectral signatures.

E. Leave-One-Sensor-Out Experiments

To further evaluate the cross-sensor generalization ability of the proposed method, we conduct leave-one-sensor-out experiments at the scale factor of ×4. Specifically, each dataset is used for testing in turn, while the other two datasets are used for training. The quantitative results are reported in Table II. It can be observed that the proposed method achieves the best performance on all five metrics under the three leave-one-sensor-out settings. When testing on the unseen Chikusei

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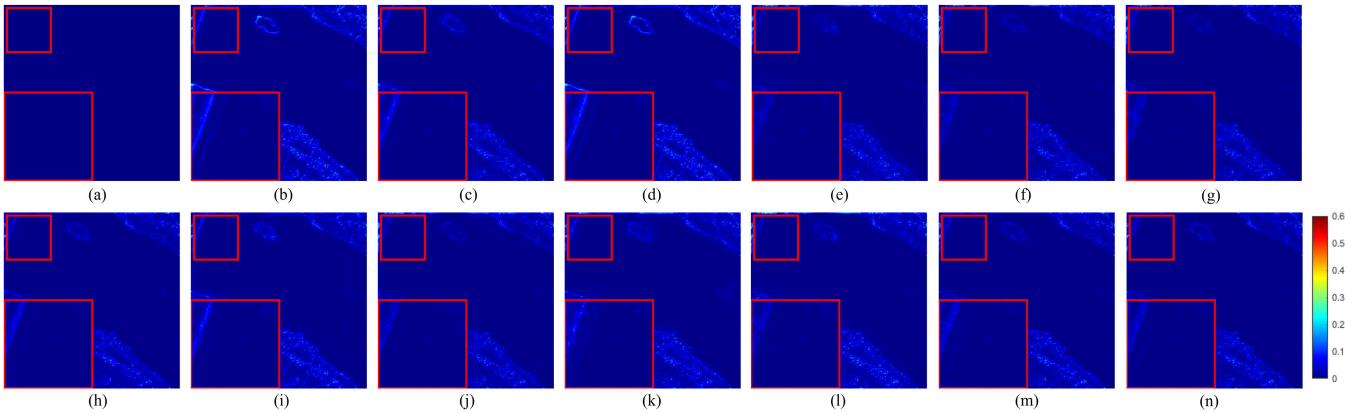


Fig. 10. Ground truth and error maps of the reconstructed HSIs on the Pavia Center dataset at a scale factor of $\times 4$. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

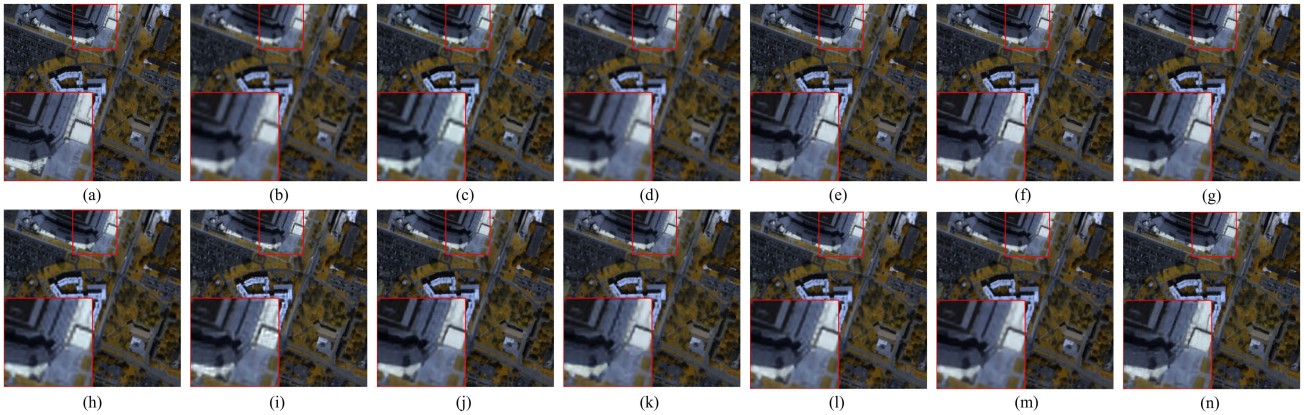


Fig. 11. Reconstructed HSI on the Houston dataset with spectral bands 29-26-19 as R-G-B with the scale factor $\times 4$. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

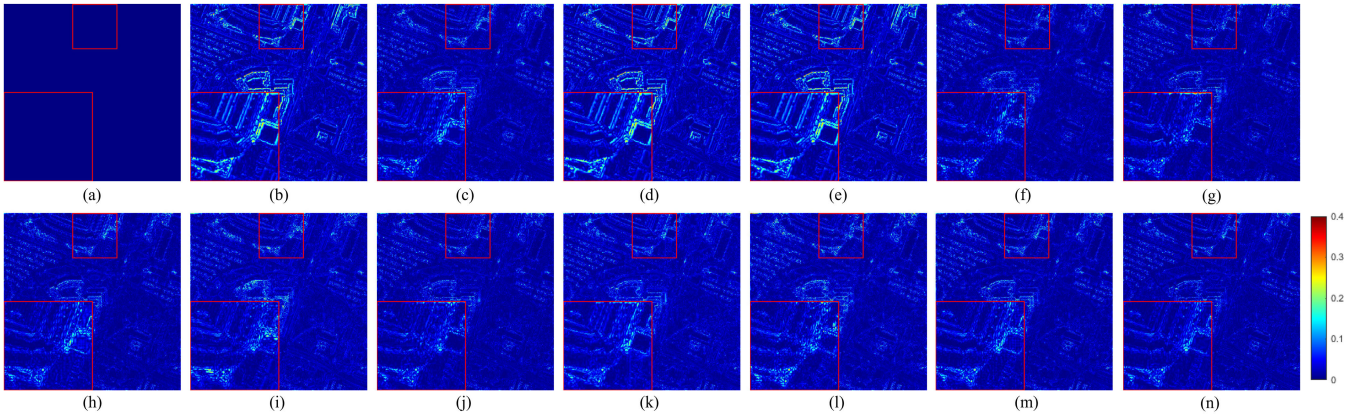


Fig. 12. Ground truth and error maps of the reconstructed HSIs on the Houston dataset at a scale factor of $\times 4$. (a) Ground truth. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

dataset, the proposed method obtains the highest PSNR of 30.7448 dB and the lowest SAM of 2.9732, outperforming the second-best method by 0.2817 dB in PSNR and 0.7589 in SAM. On the Pavia Center dataset, the proposed method also achieves the best results, with a PSNR of 28.7453 dB, SAM of 5.9745, and ERGAS of 6.5734. Compared with WESSP, the proposed method improves PSNR by 0.2327 dB and reduces ERGAS by 0.1515. When testing on the unseen Houston dataset, the proposed method achieves a PSNR of 36.2230 dB, an SAM of 1.4892, an ERGAS of 2.8992, an RMSE of 0.0198, and an SSIM of 0.9878, ranking first on all five metrics. These results show that the proposed method maintains stable reconstruction performance when applied to unseen sensor data.

F. Experiments on the Real Data

To evaluate the performance on real data, the model trained on simulated data is directly applied to nondegraded test

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TABLE II

QUANTITATIVE RESULTS OF LEAVE-ONE-SENSOR-OUT EXPERIMENTS AT A SCALE FACTOR OF 4. THE MODEL IS TRAINED ON TWO DATASETS AND TESTED ON THE REMAINING UNSEEN DATASET

	Testing on Chikusei Dataset (Cross-sensor)					Testing on Pavia Centre Dataset (Cross-sensor)					Testing on Houston Dataset (Cross-sensor)				
	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
Bicubic	28.7021	4.2947	8.0813	0.0574	0.9575	26.0185	7.5196	8.9498	0.0771	0.8562	31.2685	3.6291	4.5971	0.0344	0.9559
3DFCNN	29.9873	3.9417	6.9471	0.0477	0.9652	26.7813	6.8196	7.9498	0.0671	0.8962	32.1685	2.6291	3.7971	0.0264	0.9679
GDRRN	29.9013	4.1324	7.5127	0.0489	0.9617	26.8924	6.7758	7.8614	0.0663	0.8971	32.7249	2.5387	3.7468	0.0251	0.9688
HLNnet	30.2562	3.8869	6.4637	0.0455	0.9669	27.0316	6.7064	7.7168	0.0652	0.8993	33.2016	2.4518	3.6914	0.0236	0.9706
MCNet	30.3273	3.8125	6.2137	0.0452	0.9665	27.2148	6.6419	7.5775	0.0635	0.9021	33.5273	2.3476	3.6129	0.0237	0.9715
EUNet	30.3126	3.7853	6.1045	0.0450	0.9672	27.3897	6.5731	7.4962	0.0620	0.9078	33.8915	2.2681	3.5062	0.0239	0.9728
Interactformer	30.3785	3.7631	6.0731	0.0442	0.9676	27.6021	6.4695	7.3618	0.0607	0.9264	34.2467	2.1439	3.4286	0.0230	0.9744
SNLSR	30.3281	3.7469	6.0654	0.0448	0.9673	27.7845	6.4128	7.2036	0.0591	0.9239	34.5819	2.0574	3.3378	0.0237	0.9761
SRDNet	30.4623	3.8061	6.0973	0.0446	0.9679	28.0268	6.2964	7.1183	0.0582	0.9251	34.9735	1.9486	3.2615	0.0232	0.9785
CATANet	30.3951	3.8173	6.1052	0.0447	0.9678	28.1734	6.2389	6.9427	0.0566	0.9317	35.2148	1.8593	3.1837	0.0235	0.9807
MPHSIR	30.4327	3.7383	6.0231	0.0445	0.9680	28.4419	6.1472	6.8615	0.0548	0.9326	35.6921	1.7468	3.0649	0.0230	0.9832
WESSP	30.4631	3.7321	6.0125	0.0443	0.9681	28.5126	6.0837	6.7249	0.0541	0.9438	35.8746	1.7517	2.9814	0.0228	0.9856
Ours	**30.7448**	**2.9732**	**5.8803**	**0.0429**	**0.9698**	**28.7453**	**5.9745**	**6.5734**	**0.0523**	**0.9454**	**36.2230**	**1.4892**	**2.8992**	**0.0198**	**0.9878**

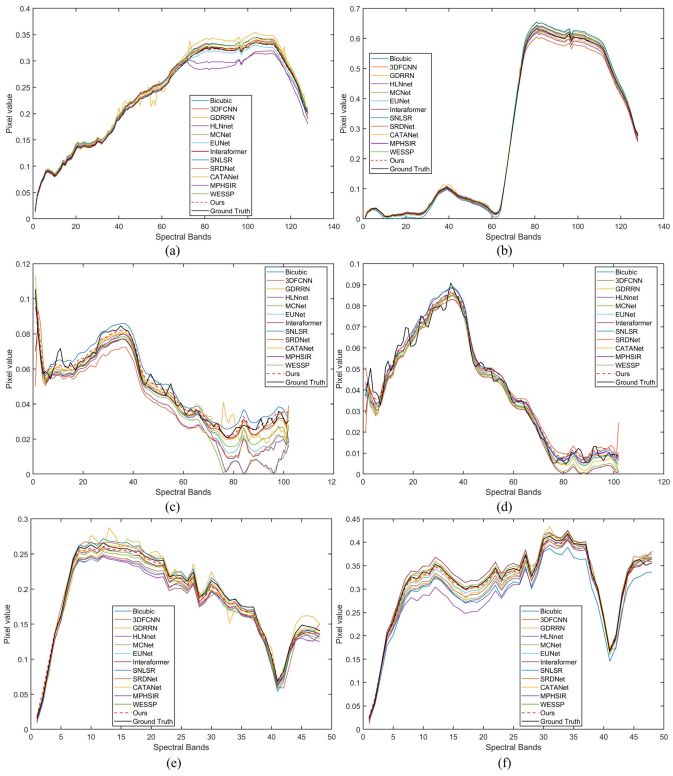


Fig. 13. Example spectral curves at selected pixels of three datasets with the scale factor of 4. (a) and (b) Chikusei dataset. (c) and (d) Pavia Center dataset. (e) and (f) Houston dataset.

TABLE III

REAL-DATA PERFORMANCE AND COMPUTATIONAL COST AT ×4

Method	Chikusei	Pavia	Houston	Params.(M)	FLOPs(G)	Time (s)
Bicubic	7.5165	6.7558	8.9867	-	-	-
3DFCNN	7.4464	7.3579	8.4633	0.04	494.50	0.05
GDRRN	8.7432	7.6437	9.1204	0.22	202.26	0.06
HLNnet	7.6347	8.4637	7.9823	9.90	8878.77	0.18
MCNet	8.1387	6.8545	8.2353	1.74	1361.52	0.12
EUNet	8.1059	7.1086	9.4563	0.62	5588.94	0.15
Interactformer	7.8773	7.5249	8.3546	4.46	5998.11	0.22
SNLSR	7.2059	7.9447	9.3454	1.65	118.43	0.08
SRDNet	9.6523	7.4976	8.4563	1.74	1361.52	0.13
CATANet	8.5671	6.4158	7.2331	0.33	389.85	0.07
MPHSIR	7.8625	6.6947	7.5732	30.91	3123.11	0.45
WESSP	6.5375	6.2518	7.3519	5.81	379.28	0.12
Ours	**6.1151**	**6.1390**	**7.1432**	0.85	2664.80	0.16

images. The visual results on the three datasets are shown in Figs. 14–16. Compared with bicubic interpolation and other deep learning-based methods, the proposed method reconstructs clearer edges and finer spatial details. In Fig. 14, our method preserves more continuous boundaries. In Fig. 15, it recovers sharper structures, such as the red roof. Fig. 16 further shows that our method produces clearer textures and better visual quality. The quantitative results are reported in Table III. The proposed method achieves the best NIQE scores on three datasets, which is consistent with the visual comparisons. Specifically, compared with the second-best results, the proposed method reduces NIQE by 0.4224, 0.1128, and 0.0899 on Chikusei, Pavia Center, and Houston datasets, respectively. These results show that the proposed method achieves better no-reference visual quality on real data in both same- and cross-sensor settings.

To evaluate the computational cost, the number of parameters (Params.), floating-point operations (FLOPs), and inference time are reported, as tabulated in Table III. The reported inference time of the proposed method includes VCA-based endmember extraction and network inference. These metrics are calculated using the Houston dataset with a scale factor of 4. Notably, the proposed method uses substantially fewer parameters than MPHSIR and WESSP. Although its FLOPs are higher than those of some methods, it achieves tradeoff between image quality and computational cost.

G. Ablation Experiments

In this section, we discuss and validate the effectiveness of the key contributions of the proposed method. For clarity, ablation experiments are conducted on three datasets with a scale factor of 2.

Ablation experiments on three datasets are conducted to verify the effectiveness of the main designs. Table IV studies the number of unfolding iterations. The performance generally improves as the number of unfolding stages increases from 1 to 5. When the number is further increased to 6, the gains become limited on several metrics. This shows that more iterations help the network refine the reconstruction, but too many iterations bring limited benefit. Therefore, we set the number of iterations to 5 in all experiments.

Table V verifies the effectiveness of EDMN and LSSSC. Removing either module generally degrades the performance across the three datasets, while the complete model achieves the best overall results. LSSSC brings a clear improvement in spectral-related metrics, especially SAM and ERGAS,

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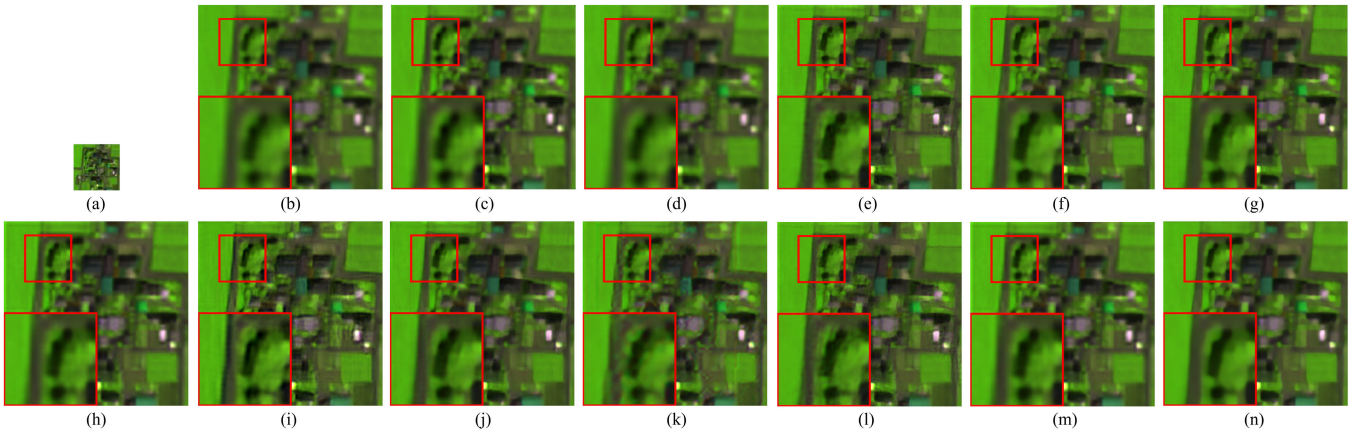


Fig. 14. Real HSI SR results in the Chikusei dataset (scale factor 4). (a) Low-resolution HSI. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

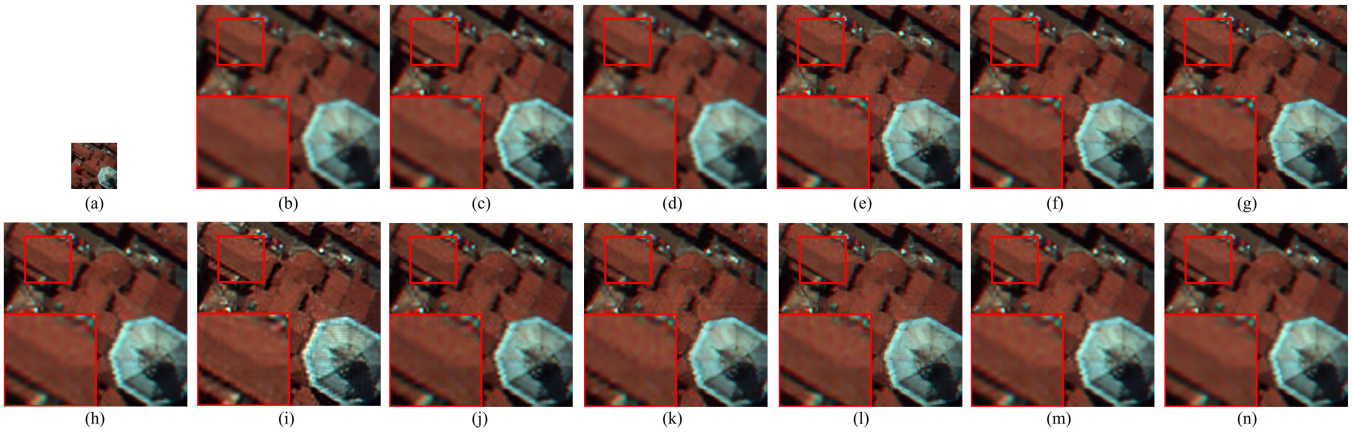


Fig. 15. Real HSI SR results in the Pavia Center dataset (scale factor 4). (a) Low-resolution HSI. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

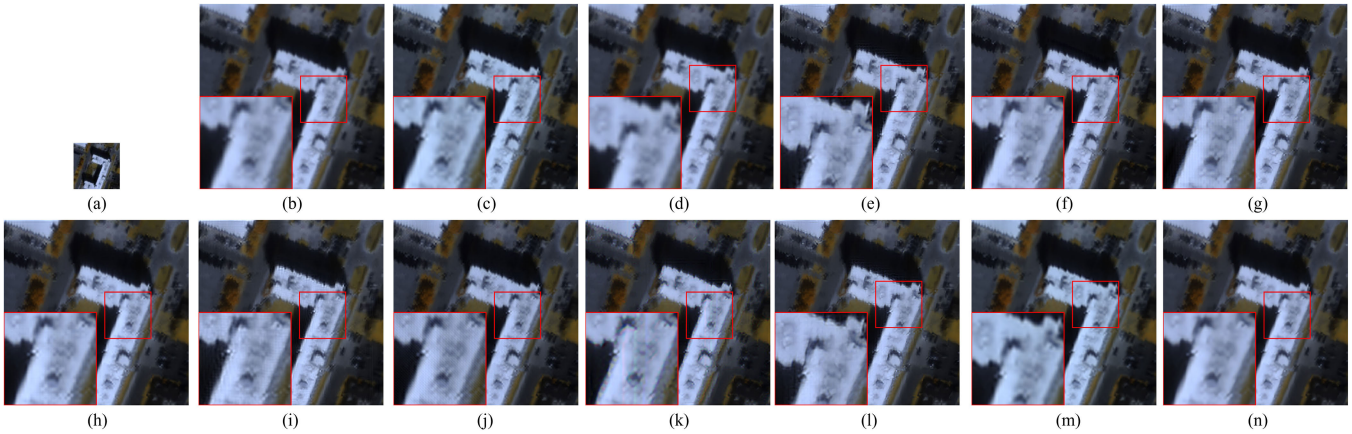


Fig. 16. Real HSI SR results in the Houston dataset (scale factor 4). (a) Low-resolution HSI. (b) Bicubic. (c) 3DFCNN. (d) GDRRN. (e) HLNnet. (f) MCNet. (g) EUNet. (h) Interactformer. (i) SNLSR. (j) SRDNet. (k) CATANet. (l) MPHSIR. (m) WESSP. (n) Ours.

which shows that the explicit spectral prior is useful for preserving spectral fidelity. EDMN also improves the results on cross-sensor datasets, indicating that transferable material-related priors are beneficial for cross-sensor generalization. Table VI further compares different variants of these two modules. Compared with both the w/o endmembers and Random endmembers variants, the proposed method achieves better performance, demonstrating the effectiveness of the transferable material-related priors from extracted endmembers. Removing either VCA or DCT reduces the performance, demonstrating the effectiveness of both endmember extraction and the DCT transformation of pooled endmember statistics in EDMN. Replacing DCT with a learnable mapping leads to lower performance across all three datasets, demonstrating that the fixed DCT transformation is more effective for processing the pooled endmember statistics in EDMN. Finally, replacing

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TABLE IV
EFFECT OF THE NUMBER OF UNFOLDING ITERATIONS

Chikusei dataset					
Unfolding stages	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
1	38.5170	1.4660	2.7228	0.0186	0.9956
2	38.5727	1.4642	2.7221	0.0185	0.9957
3	38.6605	1.4277	2.6906	0.0183	0.9958
4	38.9382	1.3797	2.6019	0.0178	0.9960
5	**39.1952**	**1.3435**	**2.5354**	**0.0172**	**0.9963**
6	39.1294	1.3678	2.5635	0.0174	0.9962
Pavia Centre dataset					
Unfolding stages	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
1	34.1964	4.3947	3.7105	0.0302	0.9809
2	34.2146	4.3840	3.7018	0.0301	0.9810
3	34.2873	4.3419	3.6813	0.0299	0.9813
4	34.2889	4.2865	3.6803	0.0298	0.9812
5	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**
6	34.4187	4.3694	3.6939	0.0295	0.9820
Houston dataset					
Unfolding stages	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
1	42.2828	1.3307	1.8481	0.0097	0.9964
2	42.3357	1.3576	1.6858	0.0096	0.9964
3	42.4797	1.2296	1.8068	0.0095	0.9966
4	42.5712	1.1890	1.3598	0.0094	0.9966
5	**42.7569**	1.1786	**1.3271**	**0.0092**	**0.9967**
6	42.7456	**1.1731**	1.3273	**0.0092**	**0.9967**

TABLE V
ABLATION STUDY OF EDMN AND LSSSC

Chikusei dataset						
EDMN	LSSSC	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
√	×	37.2958	1.7037	3.1568	0.0214	0.9942
×	√	39.0913	1.3634	2.6125	0.0179	0.9961
×	×	37.1527	1.7256	3.1984	0.0218	0.9940
√	√	**39.1952**	**1.3435**	**2.5354**	**0.0172**	**0.9963**
Pavia Centre dataset						
EDMN	LSSSC	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
√	×	33.3007	4.8566	4.1221	0.0334	0.9761
×	√	34.1453	4.3221	3.7555	0.0304	0.9804
×	×	33.1394	4.8325	4.1945	0.0341	0.9750
√	√	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**
Houston dataset						
EDMN	LSSSC	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
√	×	41.5575	1.3005	1.5414	0.0105	0.9957
×	√	42.6254	1.1966	1.3634	0.0093	0.9967
×	×	41.4431	1.3037	1.4937	0.0107	0.9957
√	√	**42.7569**	**1.1786**	**1.3271**	**0.0092**	**0.9967**

LSSSC with spectral smoothness or TV regularization leads to lower performance, demonstrating the benefit of jointly considering spectral smoothness and boundary information for spectral preservation.

Table VII analyzes the loss functions and spatial attention. Removing the spectral loss increases spectral distortion, while removing deep supervision reduces the SR performance. Replacing the spatial attention with ECA also reduces the performance, showing that spatial attention is helpful for capturing local structures in HSIs. Finally, Table VIII studies the number of endmembers. Using 20 endmembers achieves the best overall performance across the three datasets, while using 10 or 30 endmembers results in slightly lower performance. Therefore, the number of endmembers is set to 20.

TABLE VI
ABLATION STUDY OF SPECIFIC DESIGN FOR EDMN AND LSSSC

Chikusei Dataset					
Method	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
Random endmember	38.5214	1.4821	2.7845	0.0185	0.9952
w/o DCT	38.8102	1.4125	2.6512	0.0179	0.9958
w/o VCA	38.7235	1.4368	2.6892	0.0181	0.9956
w/o endmember	38.8547	1.3982	2.6243	0.0178	0.9959
Learnable mapping	38.6890	1.4423	2.7101	0.0182	0.9955
Spectral Smooth	38.9512	1.3857	2.6115	0.0177	0.9960
Standard TV	38.9861	1.3723	2.5950	0.0176	0.9961
Ours	**39.1952**	**1.3435**	**2.5354**	**0.0172**	**0.9963**
Pavia Centre Dataset					
Method	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
Random endmember	33.6211	4.5821	3.9851	0.0321	0.9802
w/o DCT	34.0523	4.4513	3.8564	0.0308	0.9809
w/o VCA	33.9856	4.4825	3.8908	0.0310	0.9807
w/o endmember	34.1230	4.4123	3.8217	0.0305	0.9811
Learnable mapping	34.0853	4.3955	3.7851	0.0299	0.9817
Spectral Smooth	34.1122	4.3213	3.7122	0.0298	0.9818
Standard TV	34.1145	4.3851	3.7543	0.0301	0.9815
Ours	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**
Houston Dataset					
Method	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
Random endmember	42.1244	1.3213	1.5853	0.0105	0.9955
w/o DCT	42.4517	1.2581	1.4825	0.0099	0.9961
w/o VCA	42.3823	1.2755	1.5124	0.0101	0.9960
w/o endmember	42.3156	1.2917	1.5456	0.0103	0.9959
Learnable mapping	42.4850	1.2188	1.4854	0.0099	0.9963
Spectral Smooth	42.5123	1.2152	1.4121	0.0098	0.9960
Standard TV	42.5219	1.2329	1.4452	0.0101	0.9961
Ours	**42.7569**	**1.1786**	**1.3271**	**0.0092**	**0.9967**

TABLE VII
ABLATION STUDY OF LOSS FUNCTIONS AND SPATIAL ATTENTION

Chikusei dataset					
	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
w/o SAM loss	39.0058	1.4527	2.5856	0.0176	0.9961
w/o deep supervision	38.7070	1.4384	2.7132	0.0183	0.9958
ECA	38.9948	1.3804	2.5938	0.0176	0.9961
Ours	**39.1952**	**1.3435**	**2.5354**	**0.0172**	**0.9963**
Pavia Centre dataset					
	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
w/o SAM loss	34.4160	4.3778	3.6214	0.0294	0.9820
w/o deep supervision	34.0982	4.4016	3.7754	0.0306	0.9805
ECA	34.4183	4.2848	3.6302	0.0294	0.9821
Ours	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**
Houston dataset					
	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
w/o SAM loss	42.5917	1.2346	1.3758	0.0093	0.9966
w/o deep supervision	42.6802	1.1965	1.4997	0.0093	0.9967
ECA	42.6055	1.2169	1.5196	0.0093	0.9966
Ours	**42.7569**	**1.1786**	**1.3271**	**0.0092**	**0.9967**

H. Classification Experiments

To further validate the performance of the cross-sensor HSI SR, the SR model trained on the Chikusei dataset is applied to the Pavia Center and Houston datasets for downstream HSI classification. For the Pavia Center dataset, the original HSI is cropped to $1080 \times 712 \times 102$ and degraded to $270 \times 178 \times 102$. The degraded images are then fed into SR models, yielding reconstructed HSIs of $1080 \times 712 \times 102$. For the Houston dataset, the original HSI is cropped to $600 \times 2384 \times 48$, degraded to $150 \times 596 \times 48$, and then reconstructed to an HSI with a size of $600 \times 2384 \times 48$.

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TABLE VIII
EFFECT OF THE NUMBER OF ENDMEMBERS

Chikusei dataset					
Number of endmembers	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
10	39.0832	1.3539	2.5626	0.0175	0.9962
20	**39.1952**	**1.3435**	**2.5354**	**0.0172**	**0.9963**
30	39.0988	1.3646	2.5708	0.0174	0.9962
Pavia Centre dataset					
Number of endmembers	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
10	34.4533	4.2825	3.6198	0.0293	0.9820
20	**34.4555**	**4.2641**	**3.6188**	**0.0293**	**0.9821**
30	34.4162	4.3778	3.6278	0.0294	0.9820
Houston dataset					
Number of endmembers	PSNR↑	SAM↓	ERGAS↓	RMSE↓	SSIM↑
10	42.6795	1.1837	1.4108	0.0093	0.9967
20	**42.7569**	**1.1786**	**1.3271**	**0.0092**	**0.9967**
30	42.7425	1.1789	1.4288	0.0092	0.9967

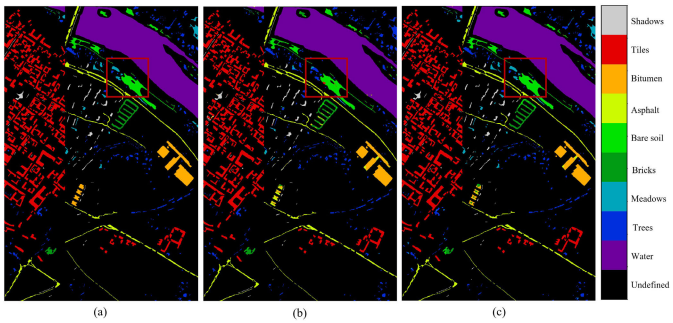


Fig. 17. Classification results of the Pavia Center dataset before and after SR. (a) Ground truth. (b) Classification result of low-resolution HSI. (c) Classification result of the reconstructed HSI.

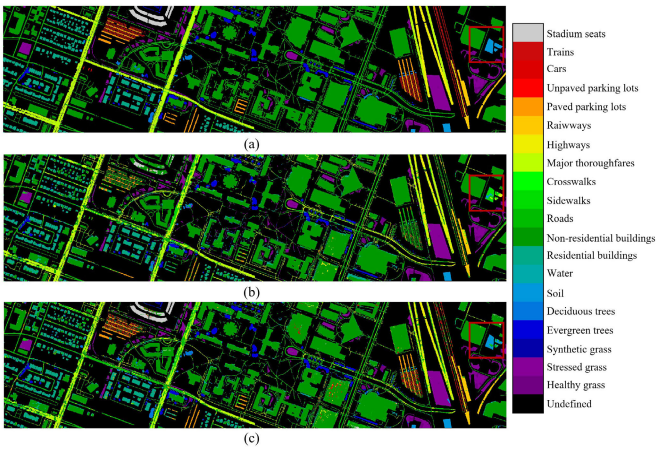


Fig. 18. Classification results of the Houston dataset before and after SR. (a) Ground truth. (b) Classification result of low-resolution HSI. (c) Classification result of the reconstructed HSI.

A support vector machine is utilized as the classifier, with 30% of the samples randomly selected from the original HSIs for the training of the support vector machine. Tables IX and X present the per-class accuracy and average accuracy (AA) for different cross-sensor datasets, where low-resolution HSIs are upsampled by a factor of 4 using bicubic interpolation. The HSIs reconstructed by the proposed method outperform those upsampled by bicubic interpolation, with AA improvements of 3.66% in the Pavia Center dataset and 9.85% in the Houston dataset. The classification maps of

TABLE IX
CLASSIFICATION RESULTS OF LOW-RESOLUTION HSI AND PREDICTED HIGH-RESOLUTION HSI ON THE PAVIA CENTER DATASET

Category	Accuracy(%)	
	LR-HSI	Predicted HR-HSI
Water	100	100
Trees	75	98
Asphalt	90	97
Self-Blocking Bricks	81	88
Bitumen	98	99
Tiles	100	100
Shadows	95	100
Meadows	90	82
Bare Soil	95	93
Average accuracy	91.56	95.22

TABLE X
CLASSIFICATION RESULTS OF LOW-RESOLUTION HSI AND PREDICTED HIGH-RESOLUTION HSI ON THE HOUSTON DATASET

Category	Accuracy(%)	
	LR-HSI	Predicted HR-HSI
Healthy grass	75	90
Stressed grass	80	91
Synthetic grass	00	96
Evergreen trees	83	84
Deciduous trees	74	71
Soil	97	87
Water	100	100
Residential buildings	59	64
Non-residential buildings	84	92
Roads	49	51
Sidewalks	40	58
Crosswalks	00	00
Major thoroughfares	53	59
Highways	61	69
Railways	86	92
Paved parking lots	62	77
Unpaved parking lots	00	00
Cars	33	76
Trains	100	84
Stadium seats	99	91
Average accuracy	61.75	71.60

the reconstructed images and the ground truth are shown in Figs. 17 and 18, indicating a high degree of similarity between the classification maps. The results of these downstream HSI classification experiments further demonstrate the performance of the proposed method in cross-sensor HSI SR.

V. CONCLUSION

In this article, we proposed an interpretable cross-sensor HSI SR framework that integrates explicit spectral priors into an optimization-driven unfolding network. An EDMN is designed to encode compact material-related information from endmembers into adaptive bandwise weights to guide iterative reconstruction, thereby improving cross-sensor generalization. Furthermore, LSSSC is incorporated into the unfolding network to improve spectral preservation. By combining explicit spectral–spatial gradient priors with adaptive attention weights, the proposed method promotes spectral smoothness in homogeneous regions while mitigating excessive smoothing near boundaries. Extensive experiments on public datasets demonstrated better overall reconstruction performance and cross-sensor generalization.

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Nevertheless, the current framework has been evaluated on only a limited number of sensors, and its performance on more diverse and real-world HSI data requires further validation. In addition, a fixed Gaussian blur kernel is used to model spatial degradation, which cannot fully represent unknown or sensor-specific point spread functions in real-world cross-sensor scenarios. The number of endmembers is empirically determined, and the endmember prior may become less reliable for scenes with different materials or spectral ranges. In future work, we will investigate sensor-specific and multiple degradation models, blind degradation estimation, dynamic endmember estimation, and unsupervised learning schemes to improve the adaptability of the proposed framework to real-world cross-sensor applications.

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